

MASSKAR, Finland

WMO: 029010

Lat: 63.728N

Lon: 22.584E

Elev: 13

StdP: 101.17

Time Zone: 2.00 (EUE)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-22.1	-27.8	0.3	-25.2	-24.2	0.4	-21.8	18.6	0.9	16.4	1.3	1.1	80	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	21.9	17.3	20.2	16.3	18.8	15.4	17.8	20.9	16.8	19.3	15.9	18.2	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	11.9	19.5	15.6	11.1	18.0	14.8	10.5	17.3	50.4	21.0	47.3	19.3	44.7	18.2	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.8	13.0	11.4	-25.6	26.0	4.5	2.4	-28.8	27.7	-31.4	29.1	-33.9	30.4	-37.2	32.1		
			-25.7	19.6	4.5	1.8	-28.9	20.9	-31.6	21.9	-34.1	22.9	-37.3	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-8.7	-7.6	-3.3	1.4	7.1	12.0	15.4	14.1	9.9	5.6	-0.1	-4.5
	DBStd	9.25	7.78	6.99	4.76	3.69	3.44	3.16	2.33	2.77	2.64	2.94	4.49	5.34
	HDD10.0	2789	578	493	413	259	104	14	0	2	34	139	302	449
	HDD18.3	5420	837	727	671	508	349	191	95	133	254	396	552	708
	CDD10.0	420	0	0	0	1	14	75	168	131	29	2	0	0
	CDD18.3	9	0	0	0	0	0	1	5	3	0	0	0	0
	CDH23.3	24	0	0	0	0	1	2	18	3	0	0	0	0
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	5.5	5.4	5.6	5.3	5.1	4.9	4.4	4.6	4.8	5.6	7.0	6.9	6.3
Precipitation	PrecAvg	528	39	27	29	26	37	50	67	57	52	56	49	43
	PrecMax	674	69	50	49	66	70	87	121	132	110	95	113	83
	PrecMin	386	10	3	5	6	5	13	5	20	20	15	5	17
	PrecStd	78	16	15	12	15	19	21	31	32	28	22	24	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.9	4.9	15.1	20.7	22.9	25.9	23.5	17.8	12.9	7.8	4.7
		MCWB	1.9	2.8	3.3	10.7	14.2	16.6	21.1	18.2	14.3	10.7	6.7	3.1
	2%	DB	2.1	3.0	3.3	10.9	16.9	20.3	22.1	21.3	15.7	11.5	6.7	3.4
		MCWB	1.0	2.1	1.8	6.7	12.3	15.5	17.8	17.4	13.5	9.9	5.8	2.5
	5%	DB	1.3	2.0	2.5	8.3	14.0	18.5	20.4	19.5	14.5	10.3	5.9	2.7
		MCWB	0.4	1.2	1.3	5.4	10.6	14.5	16.4	16.5	12.7	9.1	4.9	1.9
	10%	DB	0.5	1.0	1.8	6.4	11.8	16.7	19.0	18.1	13.5	9.5	5.1	1.9
		MCWB	-0.1	0.3	0.8	4.3	9.0	13.2	15.6	15.6	11.8	8.3	4.2	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.1	3.4	11.0	14.8	17.6	21.6	19.2	15.1	11.8	7.0	3.5
		MCDB	2.9	3.6	4.3	14.8	19.6	21.3	25.2	21.6	17.0	12.7	7.5	4.1
	2%	WB	1.3	2.2	2.2	7.2	12.5	16.2	18.2	17.9	14.0	10.1	5.9	2.7
		MCDB	1.9	2.8	2.8	10.2	16.3	19.8	21.3	20.8	15.5	11.1	6.7	3.2
	5%	WB	0.6	1.2	1.5	5.7	10.9	14.8	16.9	16.8	13.1	9.2	5.1	2.1
		MCDB	1.1	1.8	2.2	7.9	13.7	17.9	19.3	19.1	14.2	10.1	5.8	2.6
	10%	WB	-0.1	0.3	0.9	4.5	9.2	13.5	16.1	15.9	12.0	8.4	4.3	1.3
		MCDB	0.5	1.0	1.7	6.2	11.7	16.3	18.3	17.7	13.3	9.3	5.0	1.9
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.6	5.2	5.5	5.5	5.3	4.9	4.5	4.0	3.3	3.5	4.4
		MCDBR	3.6	2.9	4.4	8.3	9.0	7.6	7.2	6.8	5.2	3.7	3.0	3.2
		MCWBR	3.4	2.6	3.5	5.4	5.2	4.3	3.7	3.7	3.6	2.9	3.0	3.2
	5% WB	MCDBR	4.1	2.7	3.6	7.7	8.4	7.1	6.0	6.3	4.9	3.6	3.0	3.2
		MCWBR	4.0	2.6	3.0	5.2	5.2	4.6	3.7	3.6	3.7	3.0	3.0	3.2
Clear-Sky Solar Irradiance	taub	0.228	0.287	0.313	0.351	0.350	0.351	0.371	0.359	0.334	0.315	0.364	0.351	
	taud	2.022	2.151	2.213	2.272	2.440	2.480	2.449	2.478	2.526	2.373	2.617	2.571	
	Ebn at Noon	359	623	771	822	863	869	837	815	766	604	331	180	
	Edh at Noon	28	62	88	105	97	96	96	86	67	51	26	16	
All-Sky Solar Radiation	RadAvg	0.14	0.71	2.21	3.80	5.12	5.66	5.17	3.95	2.26	0.84	0.20	0.06	
	RadStd	0.01	0.14	0.23	0.34	0.52	0.49	0.48	0.38	0.23	0.15	0.03	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page